Dataverse Bootcamp for M365 Makers

Session downloads:
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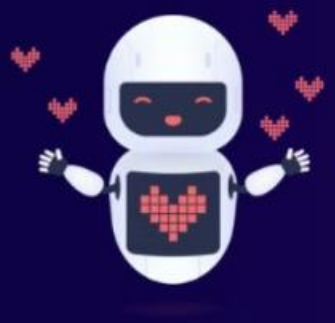

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Abstract

Dataverse, Dataverse, Dataverse... You've heard about it but haven't checked it out yet. Now's the time to see how to scale solutions up from Lists. In this hands-on workshop we'll start with a Power App built with List data and build a similar solution using Dataverse as the data source - talking through the pros and cons comparison along the way.

Adding Dataverse is an investment in time, skills, and licensing. Know when to engage and what to expect.

- Learn what Dataverse is, where it fits in the maker toolset, and how Copilot adds to the experience.
- Cover what the differences are between Dataverse, Microsoft Lists, and other data sources and talk through some of the 'what to use when' considerations as Lists solution builders integrate with and shift to Power Platform tools.
- Get a perspective on the maker skills needed when using Dataverse and how to get started in your organization.

Using the Power Apps Developer Plan, attendees will get a hands-on look at Creating Dataverse tables, relationships, forms, and more.

With Dataverse as a data source, we'll also do a quick compare of model-driven vs. canvas Power Apps.

Agenda

- What is Dataverse
- What to use when – comparing data sources
- Common organizational challenges to adoption
- Skills needed
  - Managing the platform
  - Designing and managing the data
  - Building solutions – Low code to Pro code

What is Dataverse

Microsoft Power Platform

Power BI
Business analytics

Power Apps
Application development

Power Automate
Workflow automation

Power Pages
Business websites

Microsoft Dataverse

Data Connectors

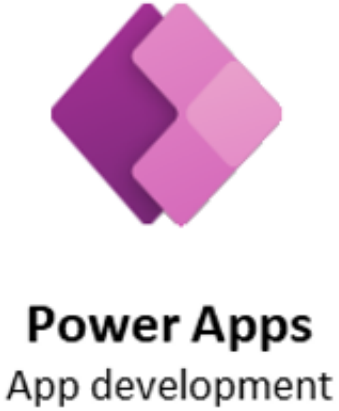
Power Fx

AI Builder

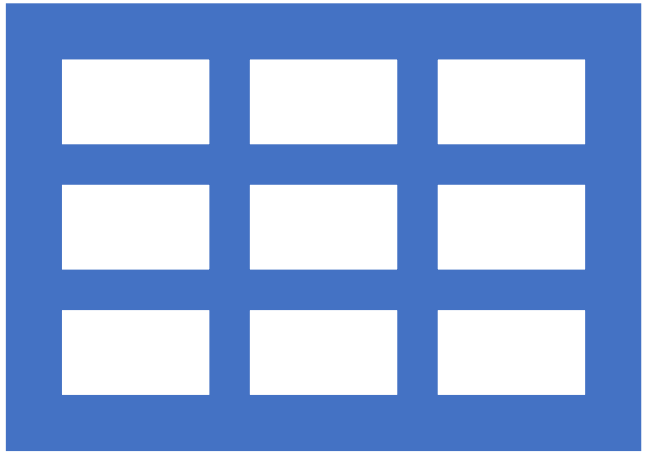
Managed Environments

Copilot Studio

App Development and Data



- Reasons to move to Dataverse
  - Larger scale, interoperability across Power Platform, and more capabilities (relational data and a lot more)
- Lists still legit
  - Better end-user experience w/data creation
  - Can scale, migrate, or integrate



Lists -> Tables

At their cores, Dataverse and Lists are tabular data

- Rows
- Columns with a variety of data types (text, numbers, dates, etc.)

You *could* take a List and build a table to do the same thing

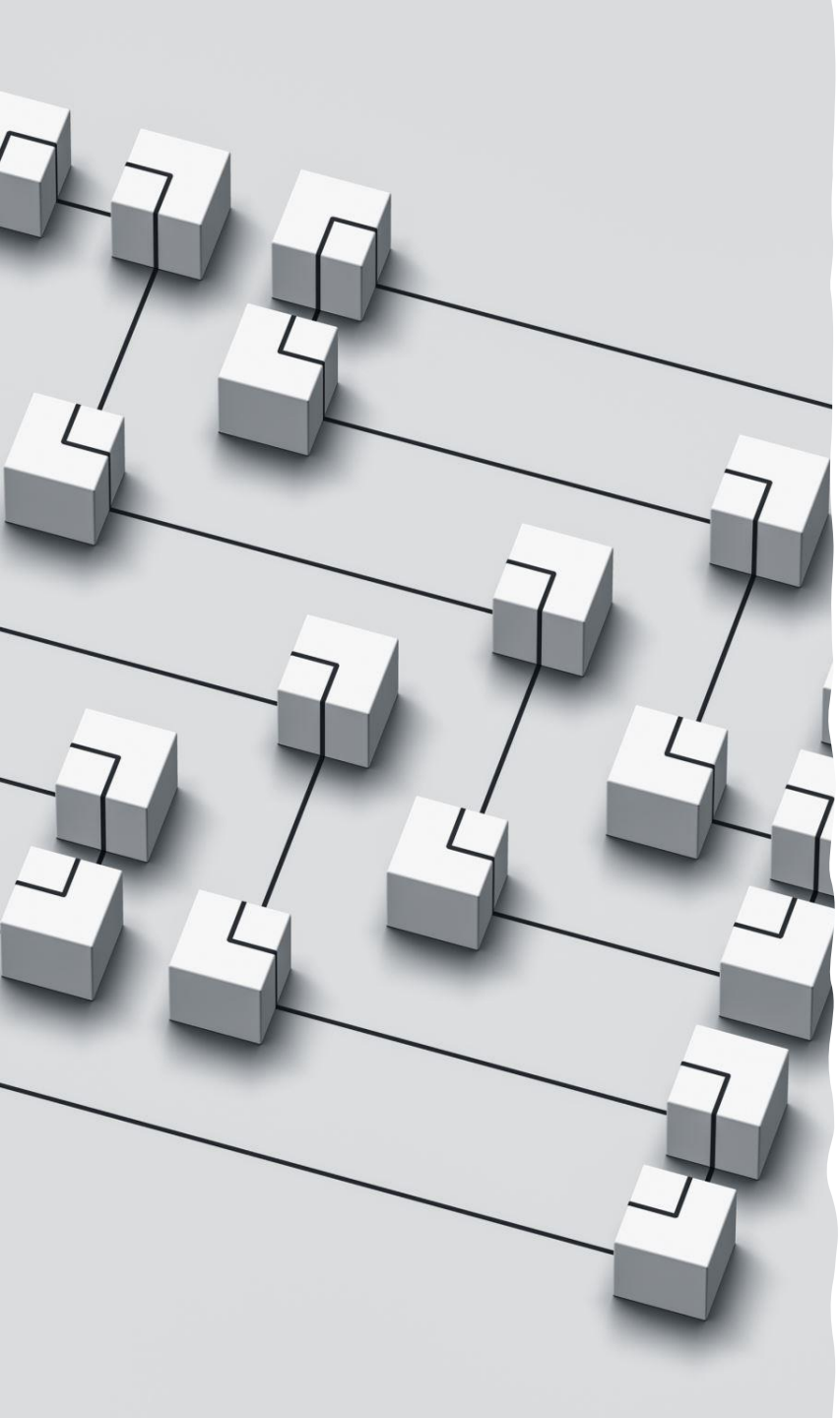
…then build more

Why consider Dataverse over Lists?

- Dataverse handles much higher volume of data
- Data relationships – Lists are NOT relational tables
- Interconnected data for multiple apps
- Performance - DV is the 1st class data store for the Power Platform

- More customizable forms
- More forms*

- Virtual Tables (not available in DV4T)
- Model Driven Apps



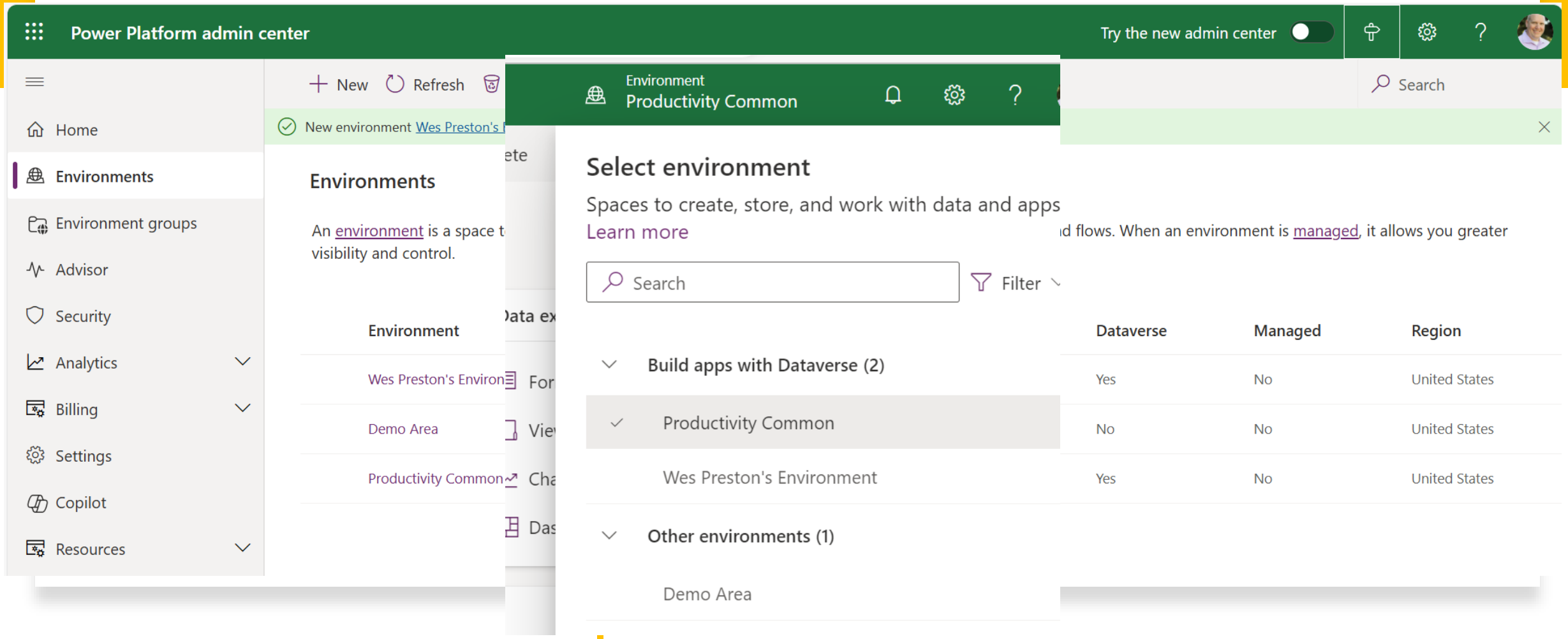
Dataverse vs. Lists - Highlights

- Security – Supports row and column-level security
- Logic – Apply at the data layer. Business rules, flows, etc.
- Data – Relational data. Allow for larger data sets.
- Storage – More Scalable
- Integration – Native connections and integration across Power Platform applications. Virtual tables from other sources.
- AI/Copilot – TBD
- ALM – Application Lifecycle Management – Dev, Stage, Prod managed environments

Microsoft Dataverse

Structure - What's in the box

Environments



Environments

- One database (grouping of tables, etc.) per environment
- Managed Environments included w/ premium licenses (not M365)
- Teams environments are special

Environments / Managed Environments

Most of us familiar with and using the Default environment (via M365)

With premium licensing -> Managed Environments

Limit sharing – Security	Weekly usage insights - Reporting	Data policies	Set up Power Platform (deployment) pipelines – ALM

Get Started – From M365

No “Dataverse” app to open

Access via other maker portals (Power Apps, etc.)

Power Platform admin center

- Https://admin.powerplatform.microsoft.com
- Lots of “Dynamics365” labeling

Power Apps maker portal -> Tables

Through Teams – Power Apps left sidebar, Templates or Build

Activity
Chat
Teams
Calendar
Calls
Files

Power Apps Home Build About

Search

All
Apps
Chatbots
Cloud flows
Component libraries
Tables
Requests
Columns

New | Edit | Using this table | Import | Export | Manage permissions | Delete

Intake Forms Demo > Tables > **Requests**

Table properties Properties Tools

Name	Primary column	Description
Requests	Name	
Type	**Last modified**	
Standard	5 hours ago	

Customizations
Columns
Relationships
Views

Power Apps Environment Productivity Common

Home
Create
Learn
Apps
Tables
Flows
Solutions
AI models

New | Edit | Create an app | Using this table | Import | Export | Advanced | Delete

Tables > **DPW Requests**

Table properties Properties Tools

Name	Primary column
DPW Requests	Title
Type	**Last modified**
Standard	1 week ago

Schema
Columns
Relationships
Keys

Data experiences
Forms
Views
Charts
Dashboards

Customizations
Business rules
Commands

Create Tables

Lots of options…

- From scratch
- Import from SharePoint List
- Use Copilot
- From a template
- Import from Access

Lab – Create Table

Dataverse Licensing

M365	Default Environment Dataverse for Teams (Limited) Power Apps and Automate to extend M365
Power Platform Apps or Automate premium plans	Per app, Per user, Pay-as-you-go plans
Dataverse Plans to add capacity	Plans to add capacity

Challenges and Approaches

Challenges to implementing Dataverse

Lack of awareness – What it is, how to implement, how to manage…

Licensing – What's included with M365 licensing is not enough

Challenged by organizational pro-dev cultures

New technology -> Additional efforts to learn

Effort to implement and manage is less than M365 as a whole, but it's new, it's more, and we're all already so busy.

Learning effort for makers is a notable jump up from Lists as a data source

Change management and/or customer success groups can help mitigate some of the effort and concerns

Licensing – It’s going to cost more

Ultimately, you need some Premium licensing

- To justify the cost, leadership needs to look at more than just one app or solution.
- Look at the speed of delivery, talking a load off of the pro-dev backlog, and the cost savings over a growing app catalog

Get Pro-dev on your side

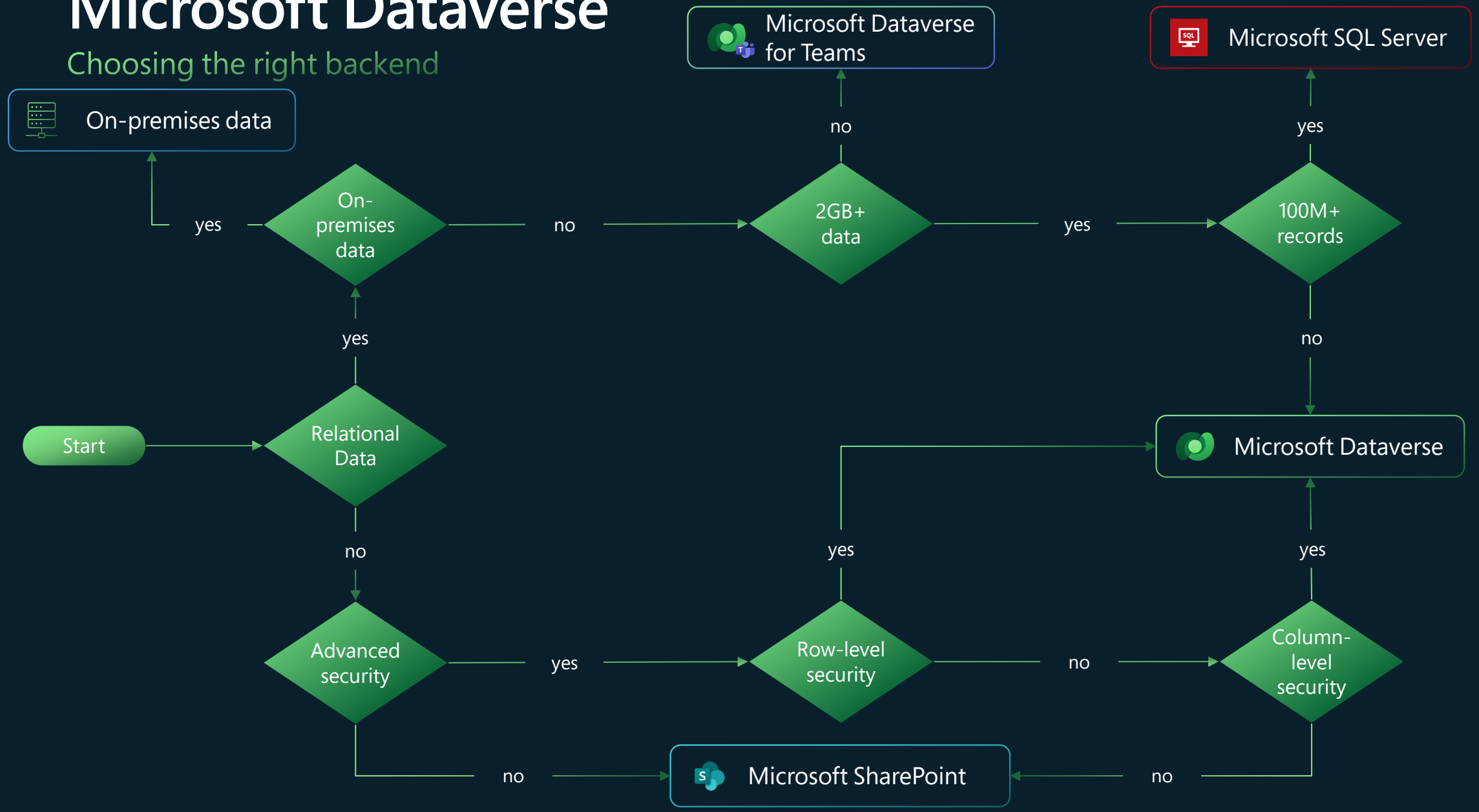
Similar effort to access APIs for List data as it is for Dataverse (if they're making custom solutions for SharePoint and Teams)

Environments, solutions, and premium licensing allow access to application lifecycle management (ALM) capabilities to deploy Dev, Stage, and Prod

What To Use When

Microsoft Dataverse

Choosing the right backend



Microsoft Dataverse and SharePoint

Choosing the right backend

Use SharePoint

1. There is a **collaborative need for the data** – e.g. Documents need to go through co-authoring.
2. **CRUD data volume is relatively small** (less than 2000 items)
3. There is **no item-level permissions** across all items. Item-level permission should be exceptions (not norm).
4. Data Access is **not exclusive to the app** and can be accessed via SharePoint's native UI as well.
5. **No requirement** for relational data
6. **Basic analytics requirement** (report and visuals on tabular data structure).

Use Dataverse

1. There is a need for data to be **created and updated frequently**.
2. Data volume requiring CRUD operations can grow **beyond 5,000 records**
3. **Advanced** security requirements are required such as row, column, hierarchy security
4. **Data Access** may need to expand beyond application to other modes or access points.
5. **Enforceable relational data** (1:N, N:1, N:N) is required
6. Solution can benefit from **analytic reporting** through Azure Data Lake.

Microsoft Dataverse

Choosing the right backend - scenarios

	SharePoint Online lists	Dataverse for Teams	Dataverse	SQL	On-premises data	Virtual Tables
Target Owner	Citizen Dev	Citizen Dev	Super user	IT Pro	IT Pro	IT Pro
Power Platform Integration	Standard Connector	Native integration with Teams	Premium connector	Premium connector	Premium connector + on-premises data gateway	Premium connector
Use Case	Low volume non-relational data	Low volume, semi-relational data in Teams	High volume transactional data	High volume master data shared across systems	Legacy master data and systems	Expose systems through Dataverse model-driven apps and pages
Feature	Single permission groups managed by super users	One-click upgrade to Dataverse	Auto load-balancing, backup, restore, audit log, advanced RBAC	Customizable data quality tools and business rules	Same gateway technology across Power Platform	CRUD and offline capabilities with model-driven apps

Power Apps Data

Lists	Dataverse Virtual Table (List, SQL Table, etc.)	Dataverse Table
• Existing data • Need to keep in Lists • Requirement for end-user involvement with data	• Existing data used by other systems or users • Can’t move • Don’t want to replicate	• Relational data • Expecting larger scope • Additional security requirements

Microsoft Lists

- Will remain a legit option for business users and citizen developers
- Still the easiest option for creating a simple data store
- Lists aren’t going anywhere
  - Continuing to improve, evolve, and introduce new capabilities, templates, and more for end users and List makers.

Skills needed

Power Platform Administrator

- Little to go on initially. No “Dataverse” Admin page.
- All Admin Centers -> Dynamics 365 Apps (Power Platform Admin center)
  - Starts in the Environments page
- Subjects to dig into:
  - Environments / Managed Environments
  - Security
  - Analytics

Maker

- Quick start – Fairly easy transition from Lists to core Table/Column set up.
- Can ramp up on newly gained (table) features over time as needed
- Can start to explore model-driven apps, Power Pages, and more

Data Analyst / Data Modeler

- When adding Dataverse as a platform, will need to work with whoever is managing data services at your organization.
- Should be able to leverage existing governance policies and extend them for Dataverse as a new platform, service, etc.
- Virtual Tables may add some new nuance to how data is managed and how connections are catalogued

References

- Working with solutions in Dataverse for Teams - Jukka Niiranen
- Create a custom table that has components with Power Apps - Power Apps | Microsoft Learn

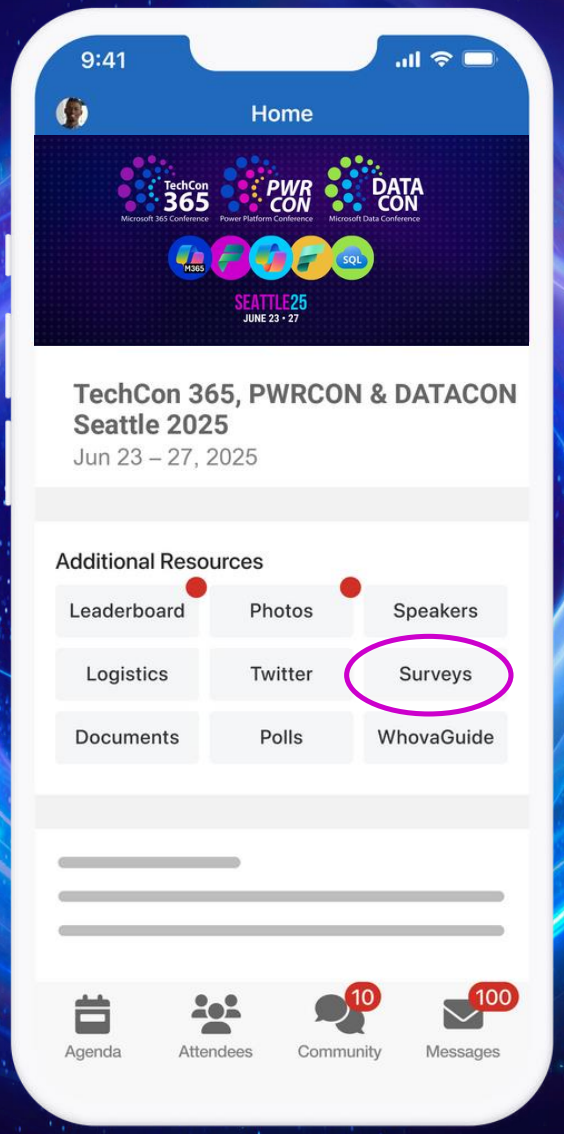
References

- About the Microsoft Dataverse for Teams environment - Power Platform | Microsoft Learn
- Dataverse for Teams vs. Dataverse - Power Apps | Microsoft Learn
- Import data from SharePoint List | Microsoft Power Apps
- About trial environments: standard and subscription-based - Power Platform | Microsoft Learn

Reminder: Trial Accounts

Don’t forget to cancel your trial account

If you do not cancel your account, you may end up with ongoing fees



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